Lab program-8:

8. Construct a C program to simulate Round Robin scheduling algorithm with C.

```c
#include <stdio.h>

int main()
{
    int i, NOP;
    int sum = 0, count = 0, y;
    int quant;
    int wt = 0, tat = 0;
    int at[10], bt[10], temp[10];

    float avg_wt, avg_tat;

    printf("Enter the Total Number of Processes: ");
    scanf("%d", &NOP);

    y = NOP;

    // Input Arrival Time and Burst Time
    for (i = 0; i < NOP; i++)
    {
        printf("\nEnter the Arrival Time and Burst Time of Process[%d]\n", i + 1);

        printf("Arrival Time : ");
        scanf("%d", &at[i]);

        printf("Burst Time   : ");
        scanf("%d", &bt[i]);

        temp[i] = bt[i];
    }

    printf("\nEnter the Time Quantum: ");
    scanf("%d", &quant);

    printf("\n----------------------------------------------------------------\n");
    printf("Process\tBurst Time\tTurnaround Time\tWaiting Time\n");
    printf("----------------------------------------------------------------\n");

    for (sum = 0, i = 0; y != 0;)
    {
        if (temp[i] <= quant && temp[i] > 0)
        {
            sum += temp[i];
            temp[i] = 0;
            count = 1;
        }
        else if (temp[i] > 0)
        {
            temp[i] -= quant;
            sum += quant;
        }

```

```c
        if (temp[i] == 0 && count == 1)
        {
            y--;

            printf("P%d\t%d\t\t%d\t\t%d\n",
                   i + 1,
                   bt[i],
                   sum - at[i],
                   sum - at[i] - bt[i]);

            wt += sum - at[i] - bt[i];
            tat += sum - at[i];

            count = 0;
        }

        if (i == NOP - 1)
        {
            i = 0;
        }
        else if (at[i + 1] <= sum)
        {
            i++;
        }
        else
        {
            i = 0;
        }
    }

    avg_wt = (float)wt / NOP;
    avg_tat = (float)tat / NOP;

    printf("----------------------------------------------------------------\n");
    printf("\nAverage Waiting Time     : %.2f", avg_wt);
    printf("\nAverage Turnaround Time : %.2f\n", avg_tat);

    return 0;
}
```

OUTPUT:

```
PS C:\Users\SMILE\Documents\OS\Lab programs> ./lab_8.exe
Enter the Total Number of Processes: 3

Enter the Arrival Time and Burst Time of Process[1]
Arrival Time : 2
Burst Time   : 5

Enter the Arrival Time and Burst Time of Process[2]
Arrival Time : 1
Burst Time   : 3

Enter the Arrival Time and Burst Time of Process[3]
Arrival Time : 2
Burst Time   : 88=

Enter the Time Quantum:
-----------------------------------------------------------------
Process Burst Time      Turnaround Time Waiting Time
-----------------------------------------------------------------
P2      3               7               4
P1      5               10              5
P3      88              94              6
-----------------------------------------------------------------

Average Waiting Time     : 5.00
Average Turnaround Time : 37.00
```